

Bin-Stable Log-Periodic Structure in Public NIST Atomic Line Lists

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Abstract

This note formalizes the method implemented in the `rickyjreyes/NIST` repository for scanning public NIST Atomic Spectra Database line-list exports in logarithmic wavenumber coordinates. The repository performs a reproducible line-density analysis, not a flux-spectrum analysis, over transition-metal line lists, with the primary canonical result reported for Fe II. For each line list, wavenumbers are cleaned, optionally filtered by ionization stage, transformed as $\ell = \ln \sigma$, binned, and compared against a Gaussian-smoothed baseline using a one-mode Poisson log-linear harmonic model. The primary observation is that Fe II yields the same best log-frequency, $k_\star = 31.3265306122449$, across 120, 160, and 200 histogram bins, with active-domain winding $n_\star \approx 10.7$ and zero null exceedances in 5000 parametric bootstrap draws for each bin setting. A separate batch run of the same scanner on Ni, Co, Cr, Mn, and Ti II line lists, also with $N_{\text{null}} = 5000$, is reported alongside the canonical Fe II result. Among the neighbors, Co II is the only species that reproduces the Fe II qualitative signature: a bin-stable $k_\star \approx 30.24$ with winding $n_\star \approx 10.95\text{--}10.98$ and zero null exceedances across all three bin counts. Ni II is statistically marginal at $N = 170$ lines; Cr II, Mn II, and Ti II do not maintain bin-stable $k_\star$ at the displayed precision. The result is therefore presented as a bin-stability observation in logarithmic line-density space, with a separate repository-inventory section distinguishing the promoted Fe II verdict-bearing result from the exploratory neighbor-ion batch and the still-pending baseline-sensitivity audits.

1 Purpose and Scope

The repository studies whether public atomic line lists contain stable harmonic structure when the spectral-line density is represented in logarithmic wavenumber coordinates. The data source is a set of public CSV exports from the NIST Atomic Spectra Database [1, 2, 3], but the analysis is not certified, endorsed, or validated by NIST. NIST is used only as the public data provider.

The implemented scan should be interpreted narrowly:

- (i) the input is a line list, not an intensity-calibrated spectrum;
- (ii) the dependent variable is line count per logarithmic-wavenumber bin;
- (iii) the fitted structure is a one-mode log-cosine modulation of a smoothed baseline;
- (iv) the main claim is bin-stability of the fitted log-frequency $k_\star$, not independent proof of a physical mechanism.

2 Relation to Log-Periodic and Discrete-Scale-Invariance Literature

Log-periodic structure is a known signature of discrete scale invariance (DSI): a system is not invariant under arbitrary continuous rescalings, but instead under a preferred discrete scaling ratio. In such cases, a leading scaling law can be decorated by oscillatory corrections periodic in the logarithm of the relevant scale. This phenomenon has been developed in the DSI literature through complex exponents, complex dimensions, hierarchical or spontaneously generated preferred scale ratios, and log-periodic corrections to scaling [4, 5, 6, 7].

The present work does not assume a power-law spectrum, a Tsallis form, or a collider cross section. It asks a narrower data-analytic question: whether public NIST atomic line lists contain a stable one-mode modulation in logarithmic wavenumber line density. The use of a log-frequency scan is therefore structurally related to the DSI/log-periodicity literature, but the observable is different. Here the dependent variable is line count per logarithmic-wavenumber bin, not transverse momentum yield, energy dissipation, rupture rate, or a multifractal moment.

This distinction matters. Prior DSI work motivates the search for stable logarithmic modes and preferred scale ratios; the present paper supplies an independent atomic-line-list test using a Poisson line-density model and a parametric bootstrap null. High-energy examples are also relevant because log-periodic modulations have been reported in transverse-momentum data/fit ratios and interpreted either through complex exponents or oscillatory scale parameters [8, 9, 10]. Those works are used here as methodological analogues, not as proof of the atomic-line result.

3 Data Cleaning and Coordinate Map

Let σ_i denote the wavenumber of the i -th usable transition line in units of cm^{-1} . The canonical scanner first parses the NIST CSV export, including spreadsheet-style cells such as `"49998.80"`, strips non-numeric flags, and extracts a positive floating-point wavenumber. If a direct wavenumber column is absent, the code may convert wavelength in nm to wavenumber using

$$\sigma_i = \frac{10^7}{\lambda_i}, \quad \lambda_i \text{ in nm}, \quad (1)$$

so that σ_i is in cm^{-1} .

The logarithmic coordinate is

$$\ell_i = \ln \sigma_i. \quad (2)$$

When an ionization-stage filter is supplied, the scanner keeps only rows whose `sp_num` field equals the requested `ion`. For the canonical Fe II result, the ion filter is `ion = 2`. After cleaning, the code sorts by wavenumber and removes duplicate wavenumber entries, leaving

$$N_{\text{FeII}} = 9447 \quad (3)$$

unique usable Fe II lines in the canonical run.

4 Histogram Line-Density Field

Fix a bin count B . The cleaned coordinates ℓ_i are binned over an active interval

$$[\ell_{\min}, \ell_{\max}], \quad \Delta\ell = \ell_{\max} - \ell_{\min}. \quad (4)$$

Let y_j be the number of spectral lines in bin j , with bin center ℓ_j . The observed line-density field is therefore

$$y_j = \#\{i : \ell_i \in \text{bin } j\}, \quad j = 1, \dots, B. \quad (5)$$

The baseline field is produced by Gaussian smoothing of the binned counts:

$$B_j = \max \{ \mathcal{G}_{\sigma_b}[y]_j, \epsilon \}, \quad (6)$$

where $\mathcal{G}_{\sigma_b}$ is a one-dimensional Gaussian filter in bin space, $\sigma_b = 6$ by default, and $\epsilon = 10^{-12}$ prevents division or logarithm singularities.

5 Poisson Deviance

For a mean model $\mu_j > 0$, the scanner uses the Poisson deviance

$$D(y||\mu) = 2 \sum_{j=1}^B \left[\mu_j - y_j + y_j \log \left(\frac{y_j}{\mu_j} \right) \right], \quad (7)$$

with the convention that the logarithmic term is omitted when $y_j = 0$. A smaller deviance indicates a better Poisson likelihood fit [12].

6 Baseline Log-Linear Fit

The smoothed field B_j is not used alone. The scanner first fits a low-order log-linear correction. Let

$$z_j = \frac{\ell_j - \bar{\ell}}{s_\ell} \quad (8)$$

when $s_\ell > 0$, and let the polynomial design matrix be

$$X_0(j, :) = \left(1, z_j, z_j^2, \dots, z_j^d \right). \quad (9)$$

The default degree is $d = 1$. The baseline model is

$$\mu_{0,j} = B_j \exp(X_0(j, :) \beta_0), \quad (10)$$

where β_0 is estimated by iteratively reweighted least squares for the Poisson log-linear model. Its deviance is denoted

$$D_0 = D(y||\mu_0). \quad (11)$$

7 One-Mode Log-Cosine Model

For each candidate log-frequency k , the code augments the baseline design with a sinusoidal pair,

$$X_k(j, :) = [X_0(j, :), \cos(k\ell_j), \sin(k\ell_j)]. \quad (12)$$

The fitted mean is

$$\mu_j(k) = B_j \exp(X_k(j, :) \beta(k) + a(k) \cos(k\ell_j) + b(k) \sin(k\ell_j)). \quad (13)$$

Equivalently, the harmonic perturbation can be written as

$$a(k) \cos(k\ell) + b(k) \sin(k\ell) = A(k) \cos(k\ell - \phi(k)), \quad (14)$$

where

$$A(k) = \sqrt{a(k)^2 + b(k)^2}, \quad \phi(k) = \text{atan2}(-b(k), a(k)). \quad (15)$$

The scan statistic at a fixed k is the deviance improvement

$$\Delta D(k) = D_0 - D_k, \quad D_k = D(y \| \mu(k)). \quad (16)$$

The observed best mode is

$$k_\star = \arg \max_{k \in \mathcal{K}} \Delta D(k), \quad \Delta D_\star = \max_{k \in \mathcal{K}} \Delta D(k). \quad (17)$$

In the canonical script, the default grid is

$$\mathcal{K} = \text{linspace}(0.5, 80.0, 2500). \quad (18)$$

8 Active-Domain Winding

The active-domain winding converts the best log-frequency into the number of oscillations across the scanned logarithmic interval:

$$n_\star = \frac{k_\star \Delta \ell}{2\pi}. \quad (19)$$

This is not an independent fit parameter. It is a derived representation of $k_\star$ over the finite active domain. The repository reports Fe II $n_\star$ values clustered near 10.7 across the three canonical bin counts.

9 Parametric Poisson Bootstrap Null

The null model is generated from the fitted baseline field μ_0 . For each bootstrap draw $r = 1, \dots, N_{\text{null}}$, synthetic counts are sampled as

$$y_j^{(r)} \sim \text{Poisson}(\mu_{0,j}). \quad (20)$$

Each synthetic dataset is scanned over the same grid $\mathcal{K}$, producing

$$\Delta D_\star^{(r)} = \max_{k \in \mathcal{K}} \left[D(y^{(r)} \| \mu_0) - D(y^{(r)} \| \mu^{(r)}(k)) \right]. \quad (21)$$

The empirical scan-level p -value is

$$\hat{p} = \frac{1 + \# \left\{ r : \Delta D_\star^{(r)} \geq \Delta D_\star \right\}}{1 + N_{\text{null}}}. \quad (22)$$

Thus, when no null draw equals or exceeds the observed improvement,

$$\hat{p} = \frac{1}{N_{\text{null}} + 1}. \quad (23)$$

For the canonical full run, $N_{\text{null}} = 5000$, so the minimum possible reported empirical value is

$$\hat{p}_{\text{min}} = \frac{1}{5001} \approx 1.9996 \times 10^{-4}. \quad (24)$$

10 Canonical Fe II Bin-Stability Result

The repository’s canonical Fe II result uses three histogram resolutions: $B \in \{120, 160, 200\}$. The reported values are:

Table 1: Canonical Fe II bin-stability result.

Bins	Unique lines	$k_\star$	$n_\star$	$\Delta D_\star$	Null tail	$\hat{p}$
120	9447	31.3265306122449	10.7171345803	355.7443010501	0/5000	0.0001999600
160	9447	31.3265306122449	10.7396495689	315.7276606037	0/5000	0.0001999600
200	9447	31.3265306122449	10.7531585620	259.1787567605	0/5000	0.0001999600

The central observation is not that $\Delta D_\star$ is identical across bin counts. It is not expected to be identical because deviance is affected by the binning resolution. The central observation is instead that the maximizer $k_\star$ is exactly stable, to the displayed precision, across all three binning choices, while the derived winding remains confined to a narrow interval near 10.7.

11 Neighbor-Ion Batch Results

The same scanner, with the same $N_{\text{null}} = 5000$ parametric Poisson bootstrap, has been run on the singly ionized line lists of five neighboring transition metals: Ti, Cr, Mn, Co, and Ni. These runs live in `outputs/batch_neighbors/` and are summarized in `outputs/batch_neighbors/batch_results.csv`. They are reported here for comparison only. As of this note, no neighbor row has been promoted into `tables/nist_master_results.csv`, so none of them is currently used by the PASS/FAIL verdict in `tables/nist_verdict.csv`.

Table 2 reports the full-null batch values for the same bin ladder $B \in \{120, 160, 200\}$ used for Fe II.

Table 2: Full $N_{\text{null}} = 5000$ neighbor-ion batch results. These rows are exploratory and do not enter the current Fe II PASS/FAIL verdict.

Species	Ion	Bins	N_{unique}	$k_\star$	$n_\star$	$\Delta D_\star$	Null tail	$\hat{p}$
Co	2	120	2450	30.2448979591837	10.9501208596	355.0073314460	0/5000	1.9996×10^{-4}
Co	2	160	2450	30.2448979591837	10.9731253152	327.1531728350	0/5000	1.9996×10^{-4}
Co	2	200	2450	30.2130852340936	10.9753715231	273.2430643153	0/5000	1.9996×10^{-4}
Ni	2	120	170	28.2725090036014	10.0617911539	25.1054218565	2/5000	6.00×10^{-4}
Ni	2	160	170	28.3997599039616	10.1283113294	24.1491909578	1/5000	4.00×10^{-4}
Ni	2	200	170	28.3679471788715	10.1296915829	20.7995400400	9/5000	2.00×10^{-3}
Cr	2	120	3898	19.1104441776711	6.9355629650	335.1167058241	0/5000	1.9996×10^{-4}
Cr	2	160	3898	19.1740696278511	6.9732729361	228.4860190362	0/5000	1.9996×10^{-4}
Cr	2	200	3898	23.7869147659064	8.6617654486	139.9934770971	0/5000	1.9996×10^{-4}
Mn	2	120	2584	14.9429771908764	5.3727910435	164.1950377932	0/5000	1.9996×10^{-4}
Mn	2	160	2584	15.1338535414166	5.4528527493	89.4593944441	0/5000	1.9996×10^{-4}
Mn	2	200	2584	23.6278511404562	8.5240189867	58.2381352468	0/5000	1.9996×10^{-4}
Ti	2	120	1578	15.9927971188475	5.6749911237	46.7594984057	0/5000	1.9996×10^{-4}
Ti	2	160	1578	66.2569027611044	23.5604355906	46.9882628356	0/5000	1.9996×10^{-4}
Ti	2	200	1578	66.2250900360144	23.5787447734	44.8162940351	0/5000	1.9996×10^{-4}

Four qualitative outcomes are visible in Table 2. Co II is the only neighbor that reproduces the Fe II signature. Co II yields $k_\star = 30.2448979591837$ at $B = 120$ and $B = 160$, and a single grid step away at $k_\star = 30.2130852340936$ for $B = 200$. The winding sits in the narrow interval

$n_\star \in [10.9501, 10.9754]$, and all three $\hat{p}$ values are at the 1/5001 floor. Co II is therefore the strongest neighbor-ion candidate for promotion, pending baseline-sensitivity and held-out audits.

Ni II is statistically marginal. The Ni II cleaned line list has only $N = 170$ usable lines. Its fitted $k_\star$ values and winding values are mutually close, but the null-tail counts are $\{2, 1, 9\}$, giving $\hat{p} \in \{6.0, 4.0, 20.0\} \times 10^{-4}$. None of these rows touches the 1/5001 floor. Ni II is therefore reported as suggestive but underpowered.

Cr II and Mn II break bin-stability between $B = 160$ and $B = 200$. Both species hold a roughly stable $k_\star$ between $B = 120$ and $B = 160$ but jump at $B = 200$. Ti II breaks bin-stability between $B = 120$ and $B = 160$, jumping from $k_\star \approx 15.99$ to $k_\star \approx 66.23$. These species reach the bootstrap floor in several rows, but their maximizing frequency is not stable across the canonical bin ladder. Thus the discriminant is not the floor p -value alone; it is bin-stability of the fitted log-mode.

12 PASS/FAIL Gate

The repository encodes the Fe II claim as a strict PASS/FAIL gate. The verdict passes only if all of the following are true:

- (1) Fe II rows exist for $B = 120, 160, 200$;
- (2) each row has 9447 unique lines;
- (3) $k_\star$ is stable across the three rows within tolerance 10^{-6} ;
- (4) $\hat{p} \leq 1/(N_{\text{null}} + 1)$, corresponding to zero null exceedances;
- (5) $10.6 \leq n_\star \leq 10.9$;
- (6) $\Delta D_\star > 0$.

The current canonical verdict is **PASS**.

13 Repository Result Inventory

The repository separates promoted, verdict-bearing results from exploratory or legacy output directories. The promoted machine-readable result set is the master CSV table `tables/nist_master_results.csv`. At the time of this note, that master table contains exactly the three canonical Fe II rows shown in Table 3. These are the only rows used by the PASS/FAIL verdict.

Table 3: Promoted rows in the repository master results table. These rows are the complete canonical result set currently used by the repository verdict.

Species	Ion	Bins	N_{unique}	$k_\star$	$n_\star$	$\Delta D_\star$	$\ell_{\min}$	$\ell_{\max}$	$\Delta \ell$	Status
Fe	2	120	9447	31.3265306122449	10.7171345803	355.7443010501	8.6610175989	10.8105611715	2.1495435726	reused
Fe	2	160	9447	31.3265306122449	10.7396495689	315.7276606037	8.6587596750	10.8128190955	2.1540594205	reused
Fe	2	200	9447	31.3265306122449	10.7531585620	259.1787567605	8.6574049207	10.8141738498	2.1567689292	reused

The verdict table `tables/nist_verdict.csv` compresses the same result into a single machine-

auditable claim row:

$$\text{claim} = \text{“Fe II log-cosine } k_\star \text{ bin-stability (bins 120/160/200)”}, \quad (25)$$

$$\text{verdict} = \text{PASS}, \quad (26)$$

$$(k_{120}, k_{160}, k_{200}) = (31.3265306122449, 31.3265306122449, 31.3265306122449), \quad (27)$$

$$(\hat{p}_{120}, \hat{p}_{160}, \hat{p}_{200}) = (0.0001999600079984003, 0.0001999600079984003, 0.0001999600079984003). \quad (28)$$

The repository also contains line-list data for neighboring transition metals and scripts for neighbor-ion batch execution. As described in the neighbor-ion section, the full $N_{\text{null}} = 5000$ batch has now been run, but those rows are not promoted into the canonical master table unless explicitly written into `tables/nist_master_results.csv`. Therefore, the paper should not imply that Ni, Co, Cr, Mn, or Ti have passed the same verdict gate as Fe II. Their status is summarized in Table 4.

Table 4: Result coverage boundary. Only the Fe II 120/160/200 bin-stability ladder is currently a promoted, verdict-bearing result.

Result class	Repository support	Claim status
Fe II bin ladder, $B = 120, 160, 200$	promoted master table	canonical PASS
Co II bin ladder, $B = 120, 160, 200$	full $N_{\text{null}} = 5000$ batch	qualitative match; promotion candidate
Ni II bin ladder, $B = 120, 160, 200$	full $N_{\text{null}} = 5000$ batch	underpowered, $N = 170$
Cr II / Mn II / Ti II bin ladders	full $N_{\text{null}} = 5000$ batch	fail bin-stability across B
Baseline-width sensitivity	partially motivated	pending audit
Alternative baselines: spline/LOESS/adaptive KDE	not promoted	pending audit
Non-parametric nulls	not promoted	pending audit
Held-out log-domain tests	not promoted	pending audit
Extended bin ladder beyond 120/160/200	not promoted	pending audit

This inventory matters because the phrase “transition-metal line lists” describes the repository input scope, whereas the phrase “canonical result” currently refers only to the Fe II promoted rows. The safe claim is therefore not that all transition metals show the mode, but rather that the promoted Fe II line-density result passes the current bin-stability gate, and Co II reproduces the qualitative signature in an exploratory batch.

14 Interpretation

The mathematical content of the result can be summarized as follows. In ordinary wavenumber space the line positions are irregularly distributed, but in logarithmic wavenumber space the Fe II line-density histogram contains a stable single-mode modulation. The fitted modulation has the form

$$\log \frac{\mu(\ell)}{\mu_0(\ell)} = c + a \cos(k_\star \ell) + b \sin(k_\star \ell), \quad (29)$$

with $k_\star = 31.3265306122449$. Across the finite active domain this corresponds to approximately

$$n_\star \approx 10.7 \quad (30)$$

cycles.

The bin-stability is important because a spurious histogram artifact would often move when the binning resolution changes. Here the reported $k_\star$ is the same across three independent histogram

resolutions. The bootstrap result is also important because the same scan procedure is applied to synthetic Poisson line-density fields drawn from the fitted smooth baseline, and no bootstrap draw reaches the observed $\Delta D_\star$ in the canonical 5000-null runs.

The neighbor-ion batch sharpens the interpretation: the Fe II signature is not unique among the scanned transition metals. Co II shows the same qualitative pattern with $k_\star \approx 30.24$, $n_\star \in [10.95, 10.98]$, bin-stability to within one grid step on the same $B \in \{120, 160, 200\}$ ladder, and zero null exceedances at $N_{\text{null}} = 5000$ at each bin count. This is not yet a verdict-level result because it has not been promoted into the master table and has not been audited for baseline-width sensitivity.

15 Limitations and Required Next Audits

This result should not be overstated. The repository inventory above shows that only the Fe II bin ladder is promoted into the verdict table. Even with the full-null neighbor-ion batch reported above, the repository itself identifies several limitations:

- (i) The analysis is a line-density scan, not a calibrated spectral-flux scan.
- (ii) NIST line-list completeness varies by species and by interval in ℓ , so the line-density field is not guaranteed to be uniformly sampled. This is especially relevant for Ni II at $N = 170$ lines.
- (iii) The null model is parametric and depends on a Gaussian-smoothed baseline.
- (iv) Baseline-width sensitivity must be audited before promoting the Co II result alongside Fe II.
- (v) The result has not yet been independently tested against alternative baseline models or non-parametric nulls.
- (vi) Cr II, Mn II, and Ti II reach the bootstrap floor in multiple rows but with bin-dependent $k_\star$. This shows that a small p -value alone is not the discriminating criterion; bin-stability is the real signature.

The strongest next checks are therefore:

- (1) replace the Gaussian-smoothed baseline with spline, LOESS, and adaptive kernel baselines, applied uniformly to Fe II and Co II;
- (2) run non-parametric nulls that preserve local line density but destroy global phase coherence;
- (3) perform held-out interval tests, fitting $k_\star$ on one part of the log-domain and testing phase/coherence on another, both for Fe II and Co II;
- (4) formally promote Co II into `tables/nist_master_results.csv` and update the verdict logic to allow a per-species PASS rather than only an Fe II-specific PASS, after audits (1)–(3) succeed;
- (5) acquire a larger Ni II line list to revisit the underpowered $N = 170$ result;
- (6) report sensitivity over σ_b , k -grid density, polynomial degree, and bin count beyond 120/160/200.

16 Conclusion

The repository implements a reproducible Poisson log-linear harmonic scan over public NIST line-list data. The canonical Fe II result is a stable log-cosine line-density mode with

$$k_{\star} = 31.3265306122449, \quad n_{\star} \approx 10.7, \quad \hat{p} = 1/5001 \quad (31)$$

across the three canonical bin counts. The appropriate claim is therefore a machine-auditable bin-stable log-density signature in Fe II. It is not, by itself, a NIST endorsement, a flux-spectrum result, or a completed physical proof. Its value is that it gives a precise, reproducible target for further null testing and physical interpretation.

A separate full- $N_{\text{null}} = 5000$ batch run on Ni, Co, Cr, Mn, and Ti II line lists shows that Co II reproduces the qualitative bin-stability signature with

$$k_{\star} \approx 30.24, \quad n_{\star} \in [10.95, 10.98], \quad \hat{p} = 1/5001 \quad (32)$$

across the same three bin counts. Co II is therefore flagged as the leading neighbor-ion promotion candidate, pending the baseline-sensitivity and held-out audits enumerated above. The other neighbors do not satisfy the bin-stability criterion at the displayed precision and are not advanced.

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